

Aligned Mathematical Structures

Source-backed grouped formula sample

1 Expressions

Expression 1. The following expression is taken from a source-backed formula corpus.

$$\begin{aligned} [M_i, p_j] &= i\epsilon_{ijk}p_k, & [M_i, p_0] &= 0, \\ [N_i, p_0] &= ip_i, & [p_\mu, p_\nu] &= 0, \\ [N_i, p_j] &= i\delta_{ij} \left[\kappa c \sinh\left(\frac{p_0}{\kappa c}\right) e^{-\frac{p_0}{\kappa c}} + \frac{1}{2\kappa c}(\vec{p})^2 \right] - \frac{i}{\kappa c} p_i p_j. \end{aligned}$$

Expression 2. The following expression is taken from a source-backed formula corpus.

$$\begin{aligned} \delta k &= \frac{1}{2}\theta k \ell, & \delta A^{(1)} &= -\theta A^{(2)}, \\ \delta e^\phi &= -\frac{7}{4}\theta \ell e^\phi, & \delta A^{(2)} &= \theta A^{(1)}, \\ \delta \ell &= \theta (1 + \ell^2 - 2k e^{-2\phi}), & \delta B &= 0, \end{aligned}$$

Expression 3. The following expression is taken from a source-backed formula corpus.

$$\begin{aligned} &|\phi_n(\alpha_a(E_\psi)) - \chi(E_\psi)| \\ &= |\phi_n(\alpha_a(E_\psi)) - \chi(E_{f_n}E_\psi)| \\ &\leq \int_{b \in U_n} f_n(b) \left(\int |\Delta(b^{-1}a)^m \psi(g_1 b^{-1}a, \dots, g_m b^{-1}a) - \psi(g_1, g_2, \dots, g_m)| dg_1 \cdots dg_m \right) db \\ &\leq \epsilon \int_{b \in U_n} f_n(b) \\ &\leq \epsilon. \end{aligned}$$

Expression 4. The following expression is taken from a source-backed formula corpus.

$$\begin{aligned} V^{(1)} &= \frac{1}{4!} f \varphi^4 + \frac{1}{48} (\beta_f - 4f\gamma_\varphi) \varphi^4 \left(\log \frac{\varphi^2}{\mu^2} - \frac{25}{6} \right) \\ &\quad - \frac{1}{2} \xi R \varphi^2 - \frac{1}{4} (\beta_\xi - 2\xi\gamma_\varphi) R \varphi^2 \left(\log \frac{\varphi^2}{\mu^2} - 3 \right), \end{aligned}$$

Expression 5. The following expression is taken from a source-backed formula corpus.

$$\begin{aligned}\xi_k^\alpha(x) &= \left(\left(\frac{1}{R(A(x))R(A(x))} \right)^{1/4} \right)^{\alpha\alpha'} B_{kk'}^{\alpha'}(x) \\ &= \frac{1}{\pi\sqrt{2}} \int_0^\infty d\lambda \lambda^{-1/4} \frac{1}{\lambda + R \bullet R} \bullet B(x)\end{aligned}$$